Answers 2.5

Part 1: Handwriting Recognition

My version	Model prediction	Correct y/n
0	7	FALSE
1	1	TRUE
2	2	TRUE
3	3	TRUE
4	4	TRUE
5	7	FALSE
6	5	FALSE
7	3	FALSE
8	2	FALSE
9	1	FALSE

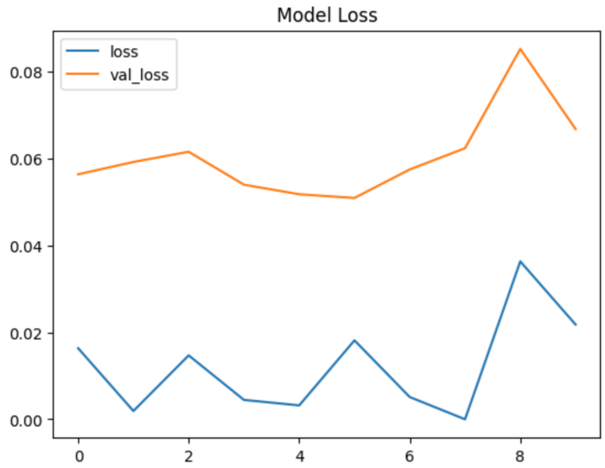
The model successfully read 40%. It was interesting to see how the predicted numbers differed, as my 7 has a horizontal line which could be interpreted as the middle line in a 3, while my 9 has the straight vertical line which could be interpreted as 1.

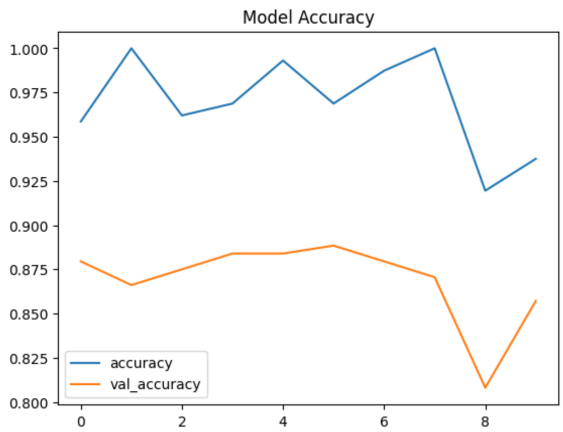
Part 2: Radar Recognition

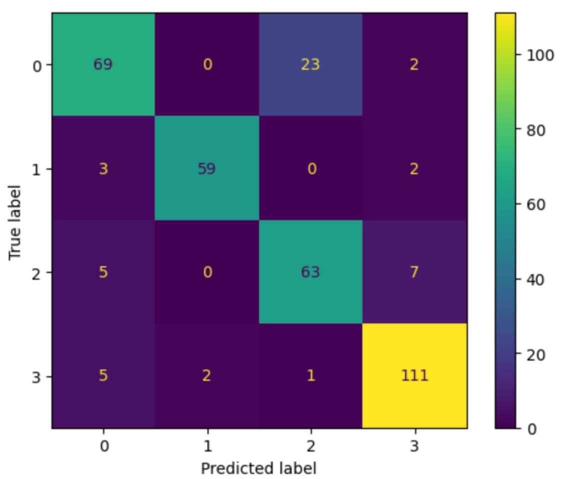
10 epochs

`Accuracy: 0.9375, Val_Accuracy: 0.8571428656578064`

`Loss: 0.02178793027997017, Val_Loss: 0.06673960387706757`



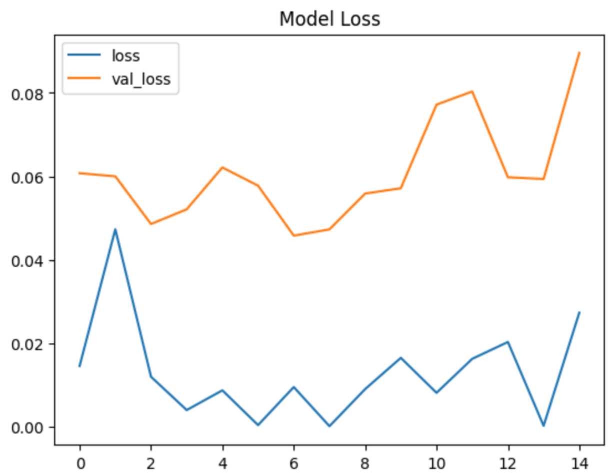


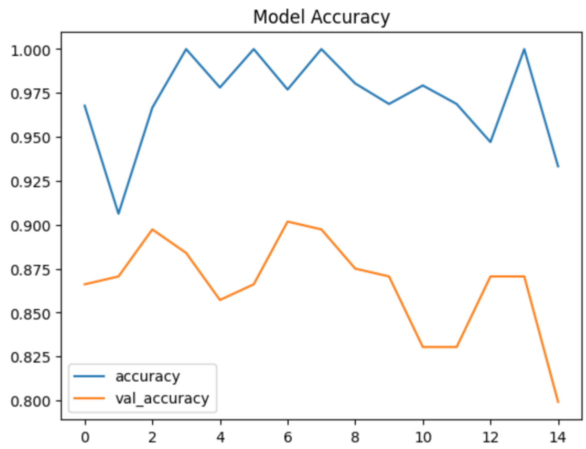


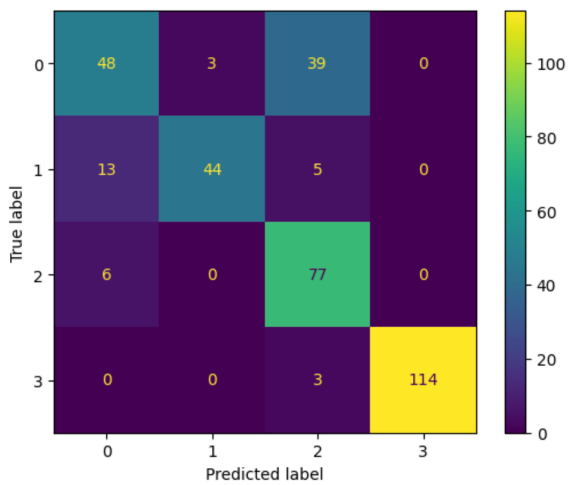
15 epochs

`Accuracy: 0.9332566261291504, Val_Accuracy: 0.7991071343421936`

`Loss: 0.02727237343788147, Val_Loss: 0.08959478884935379`



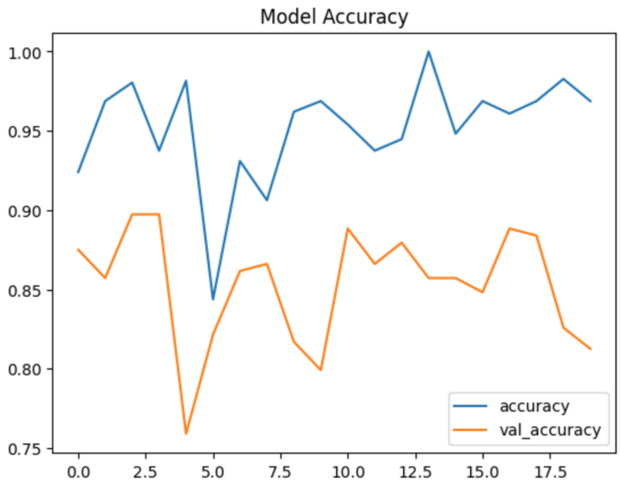


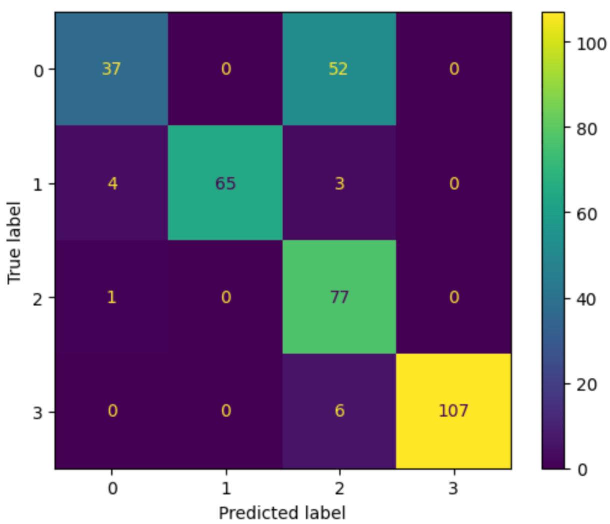


20 epochs

Accuracy: 0.96875, Val_Accuracy: 0.8125

Loss: 0.015829989686608315, Val_Loss: 0.08507144451141357





The best scores came from 10 epochs, implying that further epochs may be overfitting the data and then leading to lower matches in the confusion matrices. I had also tried higher numbers of epochs (24, 32) however these led to lower accuracies, which allowed me to narrow selection to the lower number of epochs.

The images were very interesting, especially the incorrect ones. We can see the second image was incorrect however the GAN may have put more weight on the clouds over the sunrise to assume it is 'Cloudy'. The third image has a lot of blue sky which also could be associated with its prediction of sunshine (Shine) weather.

Correct Prediction - class: Rain - predicted: Rain[9.9407143e-14 9.9999994e-01

Incorrect Prediction - class: Sunrise - predicted: Cloudy[9.9674428e-01 4.0003215e

Incorrect Prediction - class: Cloudy - predicted: Shine[1.5697266e-06 1.9604850e-

1. **Again in this same document, make a proposal for the use of GANs in weather prediction. What uses might it have?**
    - **Outline at least three ideas you have using a few sentences each.**

1. It could predict weather patterns with images for how certain weather conditions may start and then develop into others, such as a small cloud showing signs of leading to heavy rainfall. This could be useful for predicting dangerous weather conditions by analysing how they start.
2. Question an image wrongly classified as one type of weather, such as a weather pattern classified as features of a storm when it is instead likely to be moderate cloud cover (and not put off people from going outside!).
3. Create a better resolution of an image or video, so weather patterns and features can be more clearly identified. This can be useful when weather recording technology is limited or pixelated and the GAN can create a clearer image or video.
4. Create sample images of specific weather conditions for meteorologist/student training. They could range in difficulty beginning with obvious ones and progressing to those with more nuanced features, helping students identify features by eye.

- **Search for those ideas online and see if any other person or institution has researched it. If so, provide a link to the website.**

1. https://www.deeplearning.ai/the-batch/weather-forecast-by-gan/
2. https://www.preprints.org/manuscript/202306.1492
3. https://arxiv.org/html/2409.11502v1
4. https://epic.noaa.gov/2026-ams-short-course/

Upload your scripts, saved figures, and document with screenshots here for your tutor to evaluate.